



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2024-25
PRE BOARD EXAMINATION

CLASS: X
SUBJECT: ENVIRONMENTAL APPLICATIONS [SET A]

FULL MARKS: 100
TIME: 2 HOURS

*Answers to this Paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the question paper.
The time given at the head of this Paper is the time allowed for writing the answers.*

*Attempt all questions from Section A and any four questions from Section B.
The intended marks for questions or parts of questions are given in brackets [].
This question paper consists of seven printed pages.*

SECTION A (40 MARKS)
(Attempt all questions from this section)

Question 1

Choose the correct answers to the questions from the given options.
(Do not copy the questions, write the correct answers only)

[20]

- (i) Terracing is an effective technique for preventing soil erosion. What is the main purpose of terracing?
- (a) To increase water infiltration
 - (b) To prevent surface run-offs
 - (c) To reduce wind speed
 - (d) To increase biodiversity
- (ii) The main stake holder of community forestry are
- (a) State government
 - (b) Central government
 - (c) Local people
 - (d) Forest officials
- (iii)  Which of the following is a primary focus of the NGOs shown in the picture?
- (a) Promoting technological advancements for the elderly
 - (b) Providing financial support to the elderly
 - (c) Creating awareness about the importance of youth involvement in elderly care
 - (d) Protecting the rights and ensuring the well-being of senior citizens
- (iv) Which of the following is an INCORRECT statement about the sanitation of night soil?
- (a) Night soil can be safely used as fertilizer only after being properly treated or Composted.
 - (b) Direct disposal of night soil into water bodies is a major cause of water pollution.
 - (c) Night soil is completely safe when handled and stored properly.

- (d) Modern treatment systems, such as sewage treatment plants, help eliminate pathogens in night soil.
- (v) Select the appropriate option for end of the pipe treatment.
- Waste management system that processes the waste before discharging to the environment from the production unit
 - Waste management system that processes the waste after it is discharged to the environment from the production process
 - to make changes in the production process
 - meant for the product produced by a production process
- (vi) Assertion (A): Sustainable development refers to the development that meets the need of the present generation while compromising the needs of the future generation.
Reason (R): Sustainable development aims at pro-stable economic growth.
- A and R are true and R explains A
 - A and R are true but R is not the explanation of A
 - A is false but R is true
 - A is true but R is false
- (vii) Assertion (A): Micro and mini hydel power plants are not suitable for areas with seasonal variations in water flow.
Reason(R): These plants require a consistent and reliable water source to operate efficiently.
- A and R are true and R explains A
 - A and R are true but R is not the explanation of A
 - A is false but R is true
 - A is true but R is false
- (viii) When plantation activity takes place in an area where trees have been uprooted in a cyclone is known as
- Afforestation
 - Reforestation
 - Deforestation
 - Agroforestation
- (ix) Choose the odd one out in regard to fishing technique.
- Electro fishing
 - Dredging
 - Harpoon
 - Trawling
- (x) Choose the appropriate option regarding how the depletion of natural resources will affect economic development in the long run.
- It leads to increased economic growth as resources become more valuable.
 - It can constrain economic growth by inflating resource costs, unsettling industries, and contributing to social instability.
 - It has no impact on economic development, as human innovation will always find solutions.
 - It decreases economic growth as industries become less due to resource extraction.
- (a) only 1 (b) only 3 (c) 1 and 4 (d) 2 and 4

(xi) Among the statements given below select the correct ones that are true for the 'Experimental Township' at Viluppuram.

1. It encourages people from all over the world to come and live in peace and harmony.
2. It's society is based on a rigid set of rules where everyone must conform to a uniform way of life.
3. It was founded by Mirra Alfassa, known as 'The Mother', with the guidance of Sri Aurobindo.
4. It is mainly focused on urban development and industrialization to create a self-sustained economy.

(a) 4 only (b) 1 and 3 (c) 1, 3 and 4 (d) 1 and 4 only

(xii)



Choose the option that suits the picture given.

- (a) The waste is dealt off-site.
- (b) Bleaches are used in the toilet for cleaning.
- (c) The organic waste depletes soil structure and nutrition.
- (d) It is a low resource use toilet.

(xiii) Which of the following statements best reflects Dr.Rajendra Singh's philosophy about water conservation?

- (a) Only modern technology can solve water scarcity issues.
- (b) Traditional methods combined with community participation are the key to sustainable water solutions.
- (c) Government intervention is the only way to solve water problems.
- (d) Water conservation should be limited to urban areas.

(xiv) Select the correct sequence of the steps in the working of a biogas plant.

1. Feeding waste into digester
2. Anaerobic digestion
3. Collection of organic waste
4. Separation of digestate

(a) 1,3,2 and 4 (b) 3,1,2 and 4 (c) 2,4,3 and 1 (d) 4,2,1 and 3

(xv) _____ works on the principle of rotational effect and gravity.

- (a) Cyclone separators (b) Bag filters (c) Scrubbers (d) Wet scrubbers

(xvi) Which of the following is NOT a benefit of stricter emission norms like Euro and Bharat standards?

- (a) Reduced health problems caused by air pollution
- (b) Lower levels of carbon emission
- (c) Higher cost of vehicle manufacturing
- (d) Increased fuel consumption

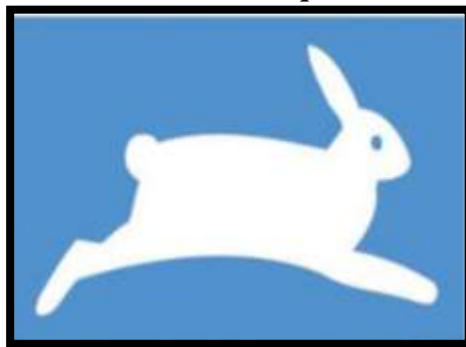
(xvii) Bagasse which is used as an alternative in paper making industries is obtained from

- (a) Walnut (b) Berries (c) Sugarcane (d) Bamboo

- (xviii) Natural resource accounting does not include
 - (a) cost of acquiring natural resources
 - (b) exporting of natural resources
 - (c) accounting of natural resource depletion
 - (d) exploration of natural resources
- (xix) Select the correct pair from the options given.
 - (a) Rural rainwater harvesting—Percolation pits
 - (b) Incineration--Leachate
 - (c) Chlorofluorocarbon---Freon
 - (d) Anna Hazare—Jal Biradari
- (xx) “If we all do something, then together there is no problem that we cannot solve!”
Rippan Kapur, founder of an NGO related to children. Identify the NGO.
 - (a) CHETNA
 - (b) CRY
 - (c) YWCA
 - (d) WWHR

Question 2

- (i) Explain Bubble Theory as an economic means to control pollution. [2]
- (ii) Study the picture given below and answer the questions that follow [2]



- (a) Identify the organization from the logo given.
- (b) What role does it play in environmental protection?
- (iii) Enumerate the various anthropogenic causes that lead to massive degradation of soil. [3]
- (iv) Discuss the environmental objectives of using incineration for waste management. [3]

Question 3

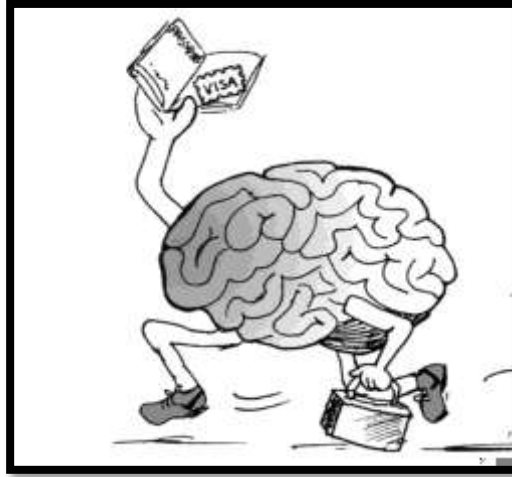
- (i) What are the key disadvantages of using landfills for waste disposal, and how do these disadvantages impact environmental sustainability and public health? [2]
- (ii) Why is it important for policymakers to monitor both GDP and GNP of a country? [2]
- (iii) List *any three* impacts Ramsar Convention will have on any wetland site. [3]
- (iv) Comment on the features of Indian Remote Sensing satellites that have made them valuable tools for environmental management. [3]

SECTION B (60 MARKS)
(Attempt any four questions from this section)

Question 4

- (i) Study the picture and answer the questions that follow:

[5]



What is being depicted in the above picture. State *any four* effects of the same.

- (ii) “If the village perishes, India will perish too. The soul of India lives in its villages.”
--Mahatma Gandhi. [5]
Why do you think Mahatma Gandhi’s vision for sustainable development in villages was holistic, community-driven and empowering?
- (iii) How can food waste be effectively repurposed to minimize environmental impact, and what are some practical applications? [5]

Question 5

- (i) After learning about water conservation in school, how would you take practical steps at home to conserve water? [5]
- (ii) What according to you are the most effective strategies for managing traffic congestion in urban areas, and how can these strategies be implemented to improve overall mobility? [5]
- (iii) “Globalization will make our societies more creative and prosperous, but also more vulnerable.”Elaborate. [5]

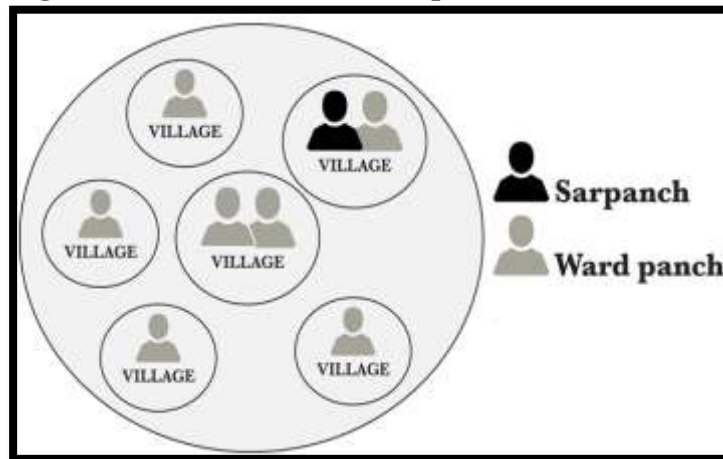
Question 6

- (i) “One who works with his hands.” Which organization is referred to here? Who is the current chairperson? What impact does this organization have on the economy and society? [5]
- (ii) Conventional sources of energy come with several significant problems. Justify the given statement. [5]
- (iii) You have learnt about various water harvesting techniques. Name and describe the one you would want to implement in your locality. [5]

Question 7

- (i) Study the picture given below and answer the question that follows

[5]



- (a) How is the above picture related to 24th April of each year?
(b) What is the importance of this organization in village communities?

- (ii) It is a government policy that any industrial project in India has to secure as a clearance from the Environment Ministry before approval for the project itself. Elaborate on the steps referred to in the above statement. [5]
- (iii) Explain how local involvement in Joint Forest Management not only supports ecological health but also fosters economic and social benefits, creating a win-win situation for both the environment and the communities that depend on it. [5]

Question 8

- (i) Decentralized wastewater treatment offers a flexible, cost-effective, and environmentally friendly alternative to traditional centralized systems. Justify the statement. [5]
- (ii) Analyze the five positive actions taken by the Agra Court to protect the three heritage sites in the Taj Trapezium Zone (TTZ), and explain how these actions contribute to the preservation of these sites. [5]
- (iii) The North-South Divide has been a source of conflict in international relations for a long time. What views do the northern and southern countries have for each other regarding environmental degradation? [5]

Question 9

- (i) Study the diagram given below and answer the question that follows:

[5]



- (a) Give an example of any NGO related to the picture given above.
(b) What roles do these NGOs play?

- (ii) **Designers are changing the perception of the textile that was once the symbol of India's freedom struggle to a sustainable fabric that belongs to the runway. What are the environmental advantages of traditional khadi weaving over modern methods?** [5]
- (iii) **A conference, 'A Sustainable Path for Biofuels', organized by Birdlife International, a conservation group, the European Environmental Bureau, a federation of European environmental organizations and European Federation for Transport and Environment, called on the European Commission to create sustainability safeguards as a part of the revision of biofuels directive.** [5]
(a) What are biofuels?
(b) How might the neglect of safeguards in biofuel production contribute to increased environmental damage and what specific consequences could arise from such oversight?